

SliderMC Command Cheat Sheet

S — Set (session)		
Short	Long	Description
SS	SetSpeed	Cruise speed mm/s ($\leq$ max_speed); bare reloads init_speed; live on next fill (incl. MJ). Dual MT: axis1=session, axis2 $\times$ ratio.
SA	SetAccel	Peak accel mm/s ² ($\leq$ max_accel); bare reloads init_accel; live on next fill (incl. MJ). Dual MT: same ratio scaling as SS.
SE	SetEnable	Driver enable 0 1; bare toggles; required before motion; off = hard stop.
ST	SetTerminal	Terminal Mode 0 1; bare toggles; local echo + UART sniff to USB (expert).
SV	SetVerbose	Verbose #... push 0 1; bare toggles; ~3 Hz (rate via verbose_rate_hz).
SD	SetDebug	USB-only debug level 0..5; bare restores default; never on UIC UART.
SL	SetLeft	Session soft min (working window); bare→slider_min; none clears (→envelope if set); skip _; !E:limit past envelope.
SR	SetRight	Session soft max; bare→slider_max; none clears; skip _; !E:limit if left>right.

G — Get (session)		
Short	Long	Description
GS	GetSpeed	Current session cruise speed.
GA	GetAccel	Current session acceleration.
GE	GetEnable	Driver enable state.
GT	GetTerminal	Terminal Mode state.
GV	GetVerbose	Verbose push state.
GD	GetDebug	USB debug level.
GL	GetLeft	Session soft min; effective (session else envelope); - if both None; dual when axis2 on.
GR	GetRight	Session soft max; same effective / - rules as GL.

I — Is / status		
Short	Long	Description
IM	IsMoving	Moving or settling on any active axis.
IH	IsHoming	Homing cycle active.
IL	IsLimit	At soft-limit position (axis1).
IE	IsError	PIN_DRV_ERROR / EMO latched.
IP	IsPosition	Axis-1 position; second field when axis2_use=1.
IA	IsAxis	Active axis count (config_axis2_enabled).
IT	IsTarget	Axis-1 seek target, or - if none / soft-stop.
IR	IsReady	1 only if idle, not homing, enabled, and not waiting.
IW	IsWaiting	1 if any W / WM / WH / WP / WC / WnC wait is active.
ID	IsDiag	Motion diag counters (FIFO underrun, peak STEP Hz, ...).
IZ	IsReset	Last chip reset: power wdt run soft debug brownout ...
IX	Pinout	ASCII GP / name / desc ($\leq$ 80 cols). Axis-2 rows only if axis2 on.

M — Movement		
Short	Long	Description
MT	MoveTo	Absolute user units; optional 2nd axis; skip _; needs SE; live-retarget. Dual: time-sync ratio.
M	Move	Relative move (alias MoveBy); same dual/skip rules as MT.
ML	MoveLeft	Jog -; mask 0=both, 1=axis1, 2=axis2 when axis2 on; soft-stop MS/!.
MR	MoveRight	Jog +; mask same as ML; soft-stop MS/!.
MJ	MoveJoy	Joy speed % of SS, signed (- left / + right); 2-axis optional pct2 (omit=0); 0=soft-stop; SS/SA live; clamp max_speed[_2].
MH	MoveHome	Homing; axis 1 (default) or 2; no-op if home_mode=0; cancel MS/H.
MS	MoveStop	Soft decelerate both axes; keeps enable; ends joy-mode; does not cancel waits. Dual: scaled accel kept.

P — Path		
Short	Long	Description
PC	PathClear	Clear path buffer (count→0); !E:busy while PG active.
PD	PathData	Append signed μ m sample(s); optional axis2; skip _ →0; OK while PG (live stream).
PG	PathGo	Play buffer from sample 0; needs SE; !E:empty busy disabled. MS/H ends path.
PN	PathNumber	Samples in buffer; allowed during PG.
PS	PathSlice	Slice length μ s ($\geq$ 1000); bare→init_path_slice_us; !E:busy while PG.

X — Extender		
Short	Long	Description
X0~3	Ext0~3	Ext out n logical 0 1; bare toggles; glued X00=X0 0; OK during EMO. X4+ rejected.
Z	Buzzer	Pulse PIN_BUZZER ~0.1 s; not a wait. No-op if BUZZER_use=0. OK during EMO/path.

C — Config		
Short	Long	Description
CS	ConfigSet	Persist key to mc.ini; silent ok. axis2_use / WDT_use need RB to take HW effect.
CR	ConfigReset	Reset all config to compiled defaults and save mc.ini.
CG	ConfigGet	One key, or bare dumps all keys (multi-line).
RB	Reboot	Soft MCU reset (no power cycle); EN off first. After CS axis2_use.

W — Wait		
Short	Long	Description
W	Wait	Delay then continue ; chain; bare→1 s; never !E:timeout.
WM	WaitMoving	Pause chain until move ends; optional timeout → !E:timeout, cancel rest of chain.
WH	WaitHoming	Pause until homing ends; timeout same as WM.
WP	WaitPos	Wait until axis-1 pos reached/overstepped; idle→immediate; 2nd arg=timeout.
WC	WaitCruise	Wait until cruise (status M) or idle; optional timeout → !E:timeout.
WnC	WaitNotCruise	Wait until not cruise M; idle/A/B→immediate; timeout same as WM.

V — Version		
Short	Long	Description
VA	VersionAbout	About string (name, version, author).
VF	VersionFW	Firmware version.
VP	VersionProtocol	Protocol version.

Special		
Short	Long	Description
H/HT	Halt	Immediate STEP abort; enable off; cancel waits and remaining ; chain.
P	Pins	Machine-readable pin map (alias VersionGPIO). Axis-2 pins if axis2 on.
\$/HL	Help	ASCII table of all commands ($\leq$ 80 columns).

Wire: one command per line (`\n`); chain with `;`; `#` comment to EOL (comment-only lines ignored); bare bool setters toggle; motion/settings silent on success; errors `!E:code message`.

Realtime (no newline): `? status` · `! soft stop` · `Ctrl-X (0x18) soft reset`.

Status (`#X ...`): `I` idle · `A` accel · `M` cruise · `B` decel · `H` homing · `P` path · `L` hard-limit · `D` disabled · `E` error. Moving: `#M/#A/#B pos speed accel [target]`.

Halt vs Stop: `MS/!` soft decel (enable kept, ends joy-mode); `H/HT` immediate abort, enable off, cancel waits. `MJ`: skip if value unchanged.